

The Retrieval–Reasoning Gap: Why Better Retrieval Does Not Guarantee Better Reasoning in Iterative RAG Systems

Anugna Yakkala

School of Computer Science and Engineering,
Vellore Institute of Technology, Vellore, TN, India

Abstract

Typical evaluations conducted using RAG evaluate the retrieval itself and infer that any improvements in retrieval will provide an improvement in the performance of the end-use task; however, we have demonstrated this to be incorrect for multi-hop reasoning tasks. Specifically, we have established the *retrieval–reasoning gap* that exists between these two components: improvements made in metrics on the retrieval side do not correlate with improved accuracy of reasoning performance when the language model is fixed. To investigate this further, an ablation study with six configurations on a set of 65 multi-hop queries created from synthetic telecom logs spanning a total of 1,115 records has been performed. Within this analysis, we found that the Precision@K values for this ablation set would vary by an average of 0.24 ($p < 0.001$), but that the root-cause accuracy for the same dataset had no statistically significant mean change ($\Delta = 0.05$, $p = 0.08$, n.s.; Pearson $r = 0.244$). Additionally, we conducted experiments using multiple variants of the same LLM (8–70 billion parameters) and confirmed that while retrieval is impacted by model capability, the accuracy of reasoning remains constant across configurations once retrieval has been set to a particular value. To assess the diagnostics of our models against one another, we constructed a composite metric for evaluating performance based on semantic similarity, ROUGE-L, and structured scoring to capture the diagnostic capabilities of our model going beyond standard word overlap ratios ($r = 0.849$ with semantic evaluation). Our data do generalize to multimodal pipelines; any system that uses heterogeneously assembled data as inputs to a generative process after retrieving relevant evidence will have the same structural gap issues related to retrieval and reasoning.

1 Introduction

The Retrieval-Augmented Generation (RAG) systems [Lewis et al., 2020, Gao et al., 2023] are evaluated according to two families of metrics: retrieval quality (Precision@K, Recall@K, nDCG) and end-task quality (accuracy of answer, F1). Implicitly, we assume that improvements to retrieval lead to improvements to the end-task. The assumption of correlation for multi-hop retrieval-augmented generation is an important but under-studied assumption since multi-hop RAG requires the assembly of evidence from multiple sources prior to the reasoning of multiple associations by an LLM.

In this paper, we challenge our core assumption and back it with controlled empirical evidence that shows that improving retrieval on its own can be insufficient as the reader model for multi-hop is constrained at all times.

We present the **retrieval–reasoning gap**: an observation that improvements to retrieval-side metrics do not translate to improvements in the accuracy of reasoning, and that the retrieval–reasoning gap is a structural property of multi-hop retrieval-augmented generation and not a function of the size or type of dataset being evaluated.

To isolate retrieval from reasoning in our experiments, we use three approaches:

1. An **ablation study using 6 configurations** of the retrieval pipeline ($\pm$ BM25, $\pm$ cross-encoder reranking, $\pm$ iterative refinement) with the LLM model held constant (Section 5.1).
2. A **multi-LLM comparison** with fixed retrieval and only changing the reasoning model across four unique architectures of 8B–70B parameters (Section 5.2).
3. **New evaluation metrics**—semantic similarity, structured scoring, composite metric—that assess reasoning quality beyond the degree of keyword overlap (Section 5.6).

We performed our evaluation on a synthetic telecom log dataset consisting of 1,115 records, 10 log files and 10 distinct failure scenarios, and 65 evaluation queries requiring multi-hop reasoning across files. While the telecom domain serves as a controlled testbed, the evaluation phenomena we discuss—including metric disagreements, independence of failure modes, and inadequacy of aggregate scores—all apply at a general level for any RAG system performing multi-hop evidence assembly followed by LLM-based reasoning.

Our experiments were conducted in a text-based modality context, but the retrieval–reasoning gap is not modality-dependent. Multimodal retrieval systems that retrieve heterogeneous evidence across multiple modalities (image+text, video+text, structured data) face the same fundamental challenge in that retrieval of relevant support materials does not necessitate downstream multimodal generator(s) reasoning correctly with that association. Thus, our proposed evaluation framework of decomposing retrieval from reasoning applies directly to these types of multimodal systems.

Contributions.

1. **Identification that the retrieval–reasoning gap is a structural property.** Across the six ablation configurations, Precision@K showed a 0.24 variation ($p < 0.001$) while the accuracy associated with the root cause showed only a 0.05 change ($p = 0.08$, n.s.; Pearson $r = 0.244$). In addition, the multi-LLM comparison with fixed retrieval demonstrates up to a 0.103 gap in reasoning accuracy attributable solely to the capability of the model.
2. **Decomposition protocol for multi-hop RAG evaluation.** We propose: (a) ablation with fixed LLM, (b) model comparison with fixed retrieval, and (c) per-query regime classification of high/low retrieval $\times$ correct/incorrect reasoning—a framework applicable to both text-only and multimodal approaches.
3. **Composite evaluation metric.** We show that the degree of keyword overlap is a noisy measure of reasoning quality and propose a composite score (semantic similarity + ROUGE-L + structured scoring) for capturing diagnostic quality to a greater degree.
4. **Dataset release.** Synthetic telecom log dataset: 1,115 records across 10 failure scenarios and 65 annotated queries with associated ground-truth root causes.

2 Related Work

Retrieval-Augmented Generation. Retrieval-Augmented Generation (RAG), with Lewis et al. [2020] providing empirical evidence that extending LMs with retrieved passages provides large benefits with knowledge-intensive tasks, and Gao et al. [2023] summarizing extensions to various summarization, code generation, and domain-specific applications. However, the majority of RAGs utilize a single-pass architecture, inadequate for analyzing multi-file log files.

Iterative and Multi-Step Retrieval. The use of self-reflective retrieval with Self-RAG [Asai et al., 2024], the ability to utilize active learning signals in Active RAG [Jiang et al., 2023], and the use of query rewriting to reformulate queries after multiple passes [Wang et al., 2023] have not provided stop criteria. Our work addresses the lack of principled stop criteria through our adaptive confidence-gated iteration, and our ablation study quantifies the benefit of iteration for both retrieval and reasoning.

Hybrid Retrieval and Cross-Encoder Reranking. Using BM25 [Robertson and Zaragoza, 2009] and a bi-encoder retrieval system [Reimers and Gurevych, 2019] is considered a standard practice in the RAG community. Studies show that using a cross-encoder reranker [Nogueira and Cho, 2019] significantly improves the precision of retrieved records and BEIR [Thakur et al., 2021] has confirmed the ability of a cross-encoder reranker to provide zero-shot generalization. Our analysis shows that while reranking does provide substantial benefits to precision of retrieval, it does not proportionally affect the accuracy of reasoning.

RAG Evaluation. The RAGAS [Es et al., 2024] and ARES [Saad-Falcon et al., 2024] evaluation frameworks decompose the quality of RAGs into three parts: context relevance, faithfulness to the source, and relevance of the answer. Our work contributes to this literature by illustrating that retrieval and end-task metrics move independently of one another, and neither metric alone or together can provide evidence as to which of the reported failure modes was responsible for the decline in performance.

3 Method

3.1 Pipeline Architecture

The four-stage iterative feedback loop consists of: *Hybrid Candidate Generation* $\rightarrow$ *Cross-Encoder Reranking* $\rightarrow$ *LLM Analysis* $\rightarrow$ *Confidence-Gated Iteration*.

Stage 1: Hybrid Candidate Generation. The first stage is a hybrid candidate generation system that uses a dense retriever (all-MiniLM-L6-v2 indexed in ChromaDB using 384-dimensional embeddings) and BM25 (with parameters $k_1=1.5$ and $b=0.75$) in a weighted manner: $s_{\text{hybrid}}(d) = 0.7 \cdot s_{\text{dense}}(d) + 0.3 \cdot s_{\text{BM25}}(d)$, resulting in 20 to 30 candidates for the second stage of cross-encoder reranking.

Stage 2: Cross-Encoder Reranking. The ms-marco-MiniLM-L-6-v2 model uses cross-attention to score each (query, document) pair, and the assigned score is the only way to rank the candidates, so any hybrid scoring (used in the first stage) is ignored to mitigate any return of noise from the first stage. The top 6 scores of the candidates move to the next stage for analysis.

Stage 3: LLM Analysis. The LLM (default: Llama 3.3 70B model, temperature 0) creates the structured output of root cause, severity of issue, resolution time frame, logical steps in the reasoning process, and an actionable recommendation. The final confidence that identifies which RAG iteration is to be reported is derived from a composite of: 30% of the mean score of the retrieval stage, 30% based upon the completeness of the output, 20% based upon the density of supporting evidence of the output, 20% based on the RAG iterations having similar output values.

Stage 4: Confidence-Gated Iteration. The confidence-gated RAG system has three distinct stopping rules that will end an iteration cycle: exit if the combined overall confidence is equal to or greater than 0.85, if either of the last two RAG iterations had no improvement (a plateau), or if the iteration cycle has exceeded three (maximum iterations permitted) to identify the best-scoring iteration for the RAG system.

3.2 Implementation Details

Retrieval Model: all-MiniLM-L6-v2 (22M parameters, 384-dim embeddings) indexed in ChromaDB (in-memory). Cross-Encoder Reranking Model: ms-marco-MiniLM-L-6-v2 (22M parame-

Algorithm 1 Adaptive Iterative RAG

Require: Query q , documents $\mathcal{D}$, max iters M , threshold ϵ

```
1:  $q_1 \leftarrow q$ ;  $c_0 \leftarrow 0$ ; best  $\leftarrow$  NULL; seen  $\leftarrow \emptyset$ 
2: for  $i = 1$  to  $M$  do
3:    $\mathcal{C}_i \leftarrow \text{HYBRIDRETRIEVE}(q_i, \mathcal{D}, N_i)$ 
4:    $\mathcal{R}_i \leftarrow \text{RERANK}(q_i, \mathcal{C}_i, k)$ 
5:    $\mathcal{R}_i \leftarrow \text{DEDUP}(\mathcal{R}_i, \text{seen})$ ; update seen
6:    $a_i \leftarrow \text{LLM}(q_i, \mathcal{R}_i)$ 
7:    $c_i \leftarrow \text{CONFIDENCE}(a_i, \mathcal{R}_i, a_{i-1})$ 
8:   if  $c_i > c_{\text{best}}$  then best  $\leftarrow (a_i, \mathcal{R}_i, c_i)$ 
9:   end if
10:  if  $c_i \geq 0.85$  or drops  $\geq 2$  then break
11:  end if
12:   $q_{i+1} \leftarrow \text{REFINEQUERY}(q, q_i, a_i)$ 
13: end for
14: return best
```

ters, trained against the MS MARCO passage dataset). LLM (default): Llama 3.3 70B via Groq (temperature set to 0, `max_tokens=2048`). Retrieval and cross-encoder reranking occurs during each session and uses a CPU for computation to generate embeddings and perform subsequent reranking. The LLM model is called via an API. All model types used in the adaptive iterative RAG system are implemented without domain-specific fine-tuning or adjustment.

4 Experimental Setup

4.1 Dataset

We created a synthetic telecom log collection to evaluate multi-hop retrieval: there are **1,115 logs** across **10 log files** from 6 different network domains (e.g., eNodeB, CU-CP, CU-UP, RAN, Core Network, Transport). The logs represent **10 different failure scenarios** and consist of a chain of cause-and-effect relationships across 2–5 log files:

1. RRC Reconfiguration Failure — 4 log files, 13 steps
2. SCTP Transport Link Failure — 45 UEs impacted (4 log files)
3. Inter-gNB Handover Failure with PDCP Data Loss (4 log files)
4. PDU Session Denied by PCF Policy (5 log files)
5. Max RLC Retransmission Causes DRB Failure (4 log files)
6. GTP-U Path Failure on Backhaul (4 log files)
7. 5G-AKA Authentication Failure with SQN Desynchronization (4 log files)
8. Downlink Scheduler Starvation Under Load (3 log files)
9. IPsec Tunnel Failure on Midhaul (3 log files)
10. AMF Overloaded by Registration Storm (3 log files)

Each log file includes a series of logs that outline a scenario and 80–120 logs that document normal operations. The user must retrieve the logs that provide relevant evidence for their query.

Dataset Rationale. The logs for our synthetic dataset do not contain personally identifiable information (PII) or private proprietary information found in commercial-grade telecom logs (for example, subscriber-specific configuration information). A synthetic corpus has three benefits for this research: (i) The researcher has complete control over the complexity of the failures presented in a corpus (i.e., type of failure and number of causal relationships), allowing consistent evaluation of the efficacy of the various methods used to identify root causes; (ii) every query in this study has a known root cause, allowing researchers to avoid confusion when annotating the

logs; and (iii) all of the data and code will be released upon acceptance for replication of any results that are reported.

4.2 Queries

65 evaluation queries were included as follows: root-cause analysis (10), failure tracing (10), multi-hop (10), temporal (10), impact analysis (10), error code (10), cross-scenario (4), contrastive (1). Of the total number of queries submitted, 55 required cross-file reasoning to determine the correct retrieval answer. Each query was annotated for the log file(s), root-cause keyword(s), and free text describing the true nature of the logs related to the query.

4.3 Ablation Configurations

There were six configurations of the models where each model used LLM (Llama 3.3 70B):

Table 1: Ablation configurations.

Config	Dense	BM25	Rerank	Iter.
Dense-Only	✓			
BM25-Only		✓		
Hybrid	✓	✓		
Hybrid+Rerank	✓	✓	✓	
Hybrid+Iter.	✓	✓		✓
Full System	✓	✓	✓	✓

4.4 Multi-LLM Comparison

All retrievals were fixed (Hybrid+Rerank, top-6) across four models: Llama 3.3 70B, Llama 3.1 8B, Mixtral 8x7B, Gemma2 9B (all used Groq; temperature = 0).

4.5 Metrics

Retrieval: Precision@K, Recall@K, Mean Reciprocal Rank (MRR). **Reasoning:** keyword overlap (RCA_{kw}), semantic similarity, structured scores, composite score ($0.35 \cdot \text{sem} + 0.30 \cdot \text{struct} + 0.20 \cdot \text{ROUGE-L} + 0.15 \cdot \text{kw}$). **LLM-as-judge:** correct/not correct (binary).

5 Results

5.1 Study of Individual Components

There were multiple important findings for the aggregate measures (Table 2).

Table 2: Ablation results (65 queries, Llama 3.3 70B). Values produced by `run_ablation.py` and filled at run time.

Config	P@K	R@K	MRR	RCA_{kw}	Conf.	Iters
Dense-Only	0.553±0.18	0.500±0.19	0.613±0.18	0.311±0.16	0.553	1.0
BM25-Only	0.598±0.18	0.509±0.21	0.629±0.17	0.314±0.16	0.550	1.0
Hybrid	0.699±0.16	0.637±0.19	0.757±0.14	0.312±0.17	0.588	1.0
Hybrid+Rerank	0.794±0.15	0.720±0.17	0.822±0.14	0.344±0.16	0.641	1.0
Hybrid+Iter.	0.628±0.19	0.704±0.16	0.716±0.16	0.338±0.15	0.606	2.0
Full-System	0.751±0.15	0.729±0.15	0.811±0.14	0.359±0.18	0.649	2.2

Findings. The most significant finding was: (i) A cross-encoder reranking produced the largest single-component improvement: Hybrid→Hybrid+Rerank produced a change in P@K of +0.096 ($p<0.001$); however, there was only a small additional increase of +0.032 ($p=0.234$, n.s.) for RCA_{kw}. (ii) A BM25 fusion produced additional useful information to support the use of an exact error code when compared to the performance from using the Dense-Only model alone and increased the P@K from 0.553 (Dense-Only) to 0.699 (Hybrid). (iii) Iteration without any reranking will produce lower precision scores because of query drift (Hybrid+Iter. P@K=0.628 vs. Hybrid 0.699); conversely, it produced a higher recall score (0.704 vs. 0.637). (iv) In all six configurations tested, the average retrieval–reasoning gap for **P@K was 0.24** ($p<0.001$); **however, the average retrieval–reasoning gap for RCA_{kw} was only 0.05** ($p=0.08$, n.s.). Therefore, this provides direct empirical evidence for the retrieval–reasoning gap. There are scatter plots of retrieval vs. reasoning accuracy ratings available in the supplementary data section that include all 390 combinations of query×configuration.

5.2 Comparison of Multiple LLMs

Table 3: Multi-LLM results with identical retrieval. Values from `run_multi_llm_comparison.py`.

Model	RCA _{kw}	Completeness	Latency	Params
Llama-3.3-70B	0.363±0.14	0.703±0.10	4.4	70B
Mixtral-8x7B	0.300±0.12	0.674±0.09	3.8	46.7B
Gemma2-9B	0.302±0.13	0.632±0.10	2.1	9B
Llama-3.1-8B	0.260±0.12	0.607±0.12	1.9	8B

The average P@K rating was nearly the same ($P@K \approx 0.794$ for all LLMs) when retrieval using the Hybrid+Rerank model was applied to each of the four proposed models; however, the corresponding RCA_{kw} ranged from 0.260 (Llama 3.1 8B) to 0.363 (Llama 3.3 70B): a range of 0.103, which was due strictly to the model capability difference. Completeness followed the same pattern as for the variance by LLM (0.607–0.703), thereby confirming **a model capability independency basis as an influencing factor in reasoning success**. Additionally, the overall drop from the 70B→8B LLM in terms of RCA due to reasoning dropped more (0.103) than the Dense-Only→Hybrid+Rerank retrieval-driven RCA change (0.032), indicating that the total effect of retrieval optimization on reasoning quality was dominated by LLM scaling.

5.3 Correlation Between Retrieval and Reasoning

Table 4: Correlation between retrieval and reasoning across all runs. Values from `run_correlation_analysis.py`.

Metric	Pearson r	p	Spearman ρ	p
Precision@K	0.244	<0.001	0.222	<0.001
Recall@K	0.208	<0.001	0.178	0.0004
MRR	0.169	0.0008	0.143	0.0045

All correlations between retrieval metrics and reasoning accuracy metrics are weak ($|r|<0.25$, $|\rho|<0.23$) but statistically significant ($p<0.005$, which indicates that this is due to the size of the sample). The Pearson $r=0.244$ for P@K resulting in a retrieval quality variance of approximately 6% (out of a maximum 100% variance) being explained by the retrieval quality; thus **confirming the existence of a retrieval–reasoning gap for our empirical data**.

5.4 Query Failure Distributions

Table 5: Per-query regime classification.

Regime	Count	%
High retrieval, high reasoning	111	28.5
High retrieval, low reasoning	84	21.5
Low retrieval, high reasoning	84	21.5
Low retrieval, low reasoning	111	28.5

43% of queries (21.5% High P@K/Low RCA + 21.5% Low P@K/High RCA) fall into off-diagonal quadrants—direct evidence that aggregate-level metrics do not allow for characterization of multiple failure modes. Also, the symmetric distribution of the overall failure patterns between the quadrants supports the independence claim since the success of retrieval does not ensure or deny correct reasoning.

5.5 Statistical Comparison of Findings

Table 6 provides results of paired t -tests on matched sample ($n=65$) queries with bootstrapped 95% confidence intervals.

Table 6: Statistical significance of key comparisons (paired t -test, $n=65$).

Comparison	Metric	Δ	p	95% CI
Hybrid $\rightarrow$ Hybrid+Rerank	P@K	+0.096	<0.001***	[0.057, 0.134]
Hybrid $\rightarrow$ Hybrid+Rerank	RCA	+0.032	0.234	[-0.021, 0.085]
Dense-Only $\rightarrow$ Full System	P@K	+0.198	<0.001***	[0.149, 0.247]
Dense-Only $\rightarrow$ Full System	RCA	+0.048	0.083	[-0.006, 0.102]
Hybrid $\rightarrow$ Full System	P@K	+0.053	0.009**	[0.013, 0.092]
Hybrid $\rightarrow$ Full System	RCA	+0.047	0.132	[-0.014, 0.107]

Retrieval gained statistically significant improvements ($p<0.01$); reasoning differences were consistently non-significant ($p>0.08$). This observed pattern is representative of the statistical signature that demonstrates the presence of a retrieval–reasoning gap for our empirical data.

5.6 Enhanced Metric Analysis

Table 7: Metric inter-correlation. Values from `run_improved_metrics.py`.

Metric Pair	Pearson r
Keyword overlap vs. Semantic similarity	0.673
Keyword overlap vs. Composite score	0.773
Semantic sim. vs. ROUGE-L F1	0.628

The association between keyword overlap and semantic similarity is only moderate ($r=0.673$), thereby indicating this metric is most appropriate for use as a noisy surrogate measure; therefore there exists approximately 55% of the variance that does not have an experimental cause. The composite-retrieval score resulted in a higher correlation with semantic similarity ($r=0.849$), confirming the assumption that utilizing a multi-metric approach reduces the vulnerability of relying on a single metric.

5.7 Illustrative Examples

Example 1: High P@K retrieval, Low RCA. SCTP Failure (S02): The Hybrid+Rerank model retrieved all files relevant for locating data related to this case; however, the LLM was not able to identify the whole chain of sequentially affected data specifically referenced in the completion of this case.

Example 2: Low P@K retrieval, High RCA. Handover failure (S03): Although the Dense-Only model missing RAN entries produced an invalid P@K score, it produced an accurate chained reasoning for this case based on previous experiences and knowledge within the domain.

Example 3: Iteration was found to be beneficial. On failure of Authentication (S07): Initially broad first-pass retrieval; however, during query refinement, injecting the terms “SQN” and “AUSF” resulted in retrieving the proper evidence on the 2nd iteration.

Example 4: Iteration was detrimental. For scheduler starvation (S08) Hybrid+Iter.: An incorrect initial examination of the data produced drift during the query leading to the retrieval of unrelated logs from a transport log.

6 Discussion

6.1 Reasons for Retrieval-Enhanced Results Not Delivering on Reasoning

The reasoning/retrieval disconnect results from different potential failure modes that limit retrieval (i.e., the limitations of index coverage, fusion weights, and reranker accuracy) versus reasoning (i.e., the limitation of an LLM to chain multiple sources of evidence together and distinguish between relevant/not-relevant context).

The reasons for this disconnect can be summarized in three ways: (i) Precision@K is reasoning and retrieval agnostic. Therefore, a document could be deemed to be relevant based on a subject matter expert’s annotation, but would not necessarily pass through an LLM as part of its evidence chain when applying multi-source reasoning. (ii) The task of evaluating end-task accuracy does not distinguish between a correct answer that is based on LLM prior knowledge versus correct based on retrieved evidence. (iii) When aggregating results across multiple queries, the overall effectiveness of retrieval is reduced due to query drift. For example, two different queries that exhibit opposite failed retrieval states produce the same aggregated outcome; however, one query will have a high return while another will have low returns. Examples of this are illustrated in the scatter plots located in our supplementary materials (Figure 1). As indicated, of the total of 390 amounts of data collected at each query configuration, there exists very little slope in the corresponding regression ($r=0.244$), but rather a large degree of vertical separation in the variance of what a user could be presented with would indicate that regardless of what was returned as being a relevant document to the query, there is significant variability in the LLM’s reasoning capabilities.

6.2 RAG Evaluation Impact on Multi-Hop Retrieval

To properly evaluate multi-hop RAG, these evaluations should: (i) separate retrieval and reasoning from one another; (ii) hold LLMs constant while varying retrieval; (iii) provide per-query performance metrics; and (iv) provide metrics of semantic accuracies based on more than simple keyword overlap. The mentioned are also applicable to multi-hop QA, RAG for scientific integration, code assistance, and agentic retrieval.

6.3 Multi-Mode RAG Impact on Retrieval/Reasoning Gap

The retrieval vs. reasoning gap will likely be worsened when applied to multi-modal environments. When evidence of heterogeneous type (i.e., image, table, video, and written evidence) from multiple sources is retrieved, there is no guarantee that each of the multi-modal retrievals will be successfully synthesized into coherent reasoning by a generation model, even if retrieval has very high recall rates for each mode. As noted above concerning how difficult it is to effectively construct a reasoning framework from disparate modes of evidence, the multi-hop research that we are studying provides relevant multi-hop evidence types, but since each of these types are not always modularized by construction, a retrieval model must also provide a proper and aligned conceptual construct across each type of evidence. The construct comparisons between varied retrieval types and a model that performs the actual reasoning should provide a means to evaluate multi-modal generation across large numbers of iterations.

6.4 Adaptive to Fixed Iteration

The adaptive stopping sequence provides an efficient means of obtaining LLM evidence, while allowing for temporary downtimes in confidence. However, systems utilizing fixed iteration processes will have drift or change in user queries (see Example 4) and the first-pass outcome of a fixed system will affect the overall return due to the absence of precision-oriented reranking.

7 Limitations and Future Work

Text-only Evaluation. All experiments use only text as input for retrieval and generation. More work is needed to investigate whether the retrieval/reasoning gap still exists or increases when multimodal retrieval-augmented generation (RAG) is used (for example, with image+text or video+text evidence).

Synthetic Dataset. The corpus used in this study was synthetically generated, with known ground truth. While this allows for controlled assessment, there are many factors that may differ between the synthetic and true noise distributions, as well as the level of uncertainty related to the annotation and the size of the production systems. Therefore, it will be necessary to conduct more work using real-world telecommunications logs and established multi-hop benchmark datasets (e.g., HotpotQA and MuSiQue) in order to assess the generalizability of the models developed in this paper.

LLM-as-Judge Bias. The development of cross-model judges and human evaluation will provide additional support for the automated scoring methodologies.

Single Domain. Performing the same experiment using other domains (e.g., code-assistant, scientific Q&A, and medical) will strengthen the generality claim from this work.

Confidence Calibration. The composite confidence score is only a heuristic, not a formally calibrated metric.

Future work will consist of: (i) multimodal retrieval–reasoning experiments that include image+text evidence; (ii) employing larger-than-individual-size corpora with bootstrap significance testing; (iii) performing a human-calibrated evaluation of these models; (iv) adapting domain-specific embeddings created using contrastive learning methods; (v) developing graph-based systems for causal retrieval; and (vi) fine-tuning the reader models to determine whether such fine-tuning allows for advancement in reasoning performance and reduction of the retrieval/reasoning gap.

8 Conclusion

The retrieval–reasoning gap has been demonstrated in this work: across six different ablation configurations and the sixty-five queries, P@K varied 0.24 ($p<0.001$) while RCA_{kw} only varied by

0.05 ($p=0.08$, n.s.) with Pearson $r=0.244$ between retrieval and reasoning. In our investigation of multiple LLMs with fixed retrieval and their corresponding reasoning accuracies, we show that, at times, simply because of the model capability they displayed there was a reasoning accuracy difference of as much as 0.103. Of the forty-three percent of our per-query analysis, we found all cases fall into the off-diagonal area of the retrieval–reasoning space indicating that retrieval and reasoning do not agree upon the same solution. We also introduced an approach to break down (decompose) the LLMs to create a single evaluation (composite metric) that showed good agreement ($r=0.849$ with semantic evaluation) with the original semantic evaluation data. Therefore, the implication is clear: though both retrieval and reasoning improvements must be pursued, improving retrieval alone is not enough for multi-hop RAG success. Thus, reasoning quality must be independently addressed and also measured during the evaluation process of RAGs.

References

- Asai, A., Wu, Z., Wang, Y., Sil, A., and Hajishirzi, H. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*, 2024.
- Gao, Y., Xiong, Y., Dibia, V., et al. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.
- Jiang, Z., Xu, F. F., Gao, L., et al. Active retrieval augmented generation. In *EMNLP*, 2023.
- Khattab, O. and Zaharia, M. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *SIGIR*, 2020.
- Lewis, P., Perez, E., Piktus, A., et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, 2020.
- Nguyen, T., Rosenberg, M., Song, X., et al. MS MARCO: A human generated machine reading comprehension dataset. In *NeurIPS Workshop on Cognitive Computation*, 2016.
- Nogueira, R. and Cho, K. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.
- Reimers, N. and Gurevych, I. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *EMNLP*, 2019.
- Robertson, S. and Zaragoza, H. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in IR*, 3(4):333–389, 2009.
- Thakur, N., Reimers, N., et al. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *NeurIPS Datasets and Benchmarks*, 2021.
- Wang, L., Yang, N., and Wei, F. Query2doc: Query expansion with large language models. In *EMNLP*, 2023.
- Es, S., James, J., Espinosa-Anke, L., and Schockaert, S. RAGAS: Automated evaluation of retrieval augmented generation. In *EACL (System Demonstrations)*, 2024.
- Saad-Falcon, J., Khattab, O., Potts, C., and Zaharia, M. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *NAACL*, 2024.
- Upadhyay, S., Pradeep, R., Thakur, N., et al. A large-scale study of relevance assessments with large language models: An initial look at the TREC 2024 RAG Track. *arXiv preprint arXiv:2411.08275*, 2024.

Liu, N. F., Lin, K., Hewitt, J., et al. Lost in the middle: How language models use long contexts. *TACL*, 12:157–173, 2024.